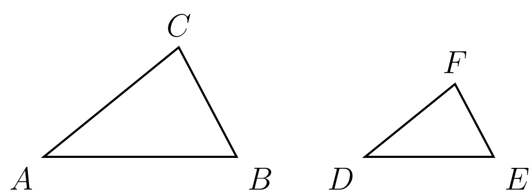
Triangle Similarity — Diagnosing Reversed Correspondence and Scale-Factor Errors

High School Geometry

In a similarity statement the **order of the letters is the correspondence**: $\triangle ABC \sim \triangle DEF$ says $A \leftrightarrow D$, $B \leftrightarrow E$, $C \leftrightarrow F$, so $\overline{AB}$ pairs with $\overline{DE}$ and $\overline{BC}$ with $\overline{EF}$.

- A **reversed correspondence** pairs a side with the wrong partner.
- A **reversed scale factor** uses large-over-small where small-over-large belongs (or the reverse), so a side that should grow shrinks instead.
- Before computing, decide which triangle is larger; then say whether your answer should be bigger or smaller than the side you started from.



How every similar pair on this sheet is named: matching letters mark matching vertices. No values shown — use the numbers given in each problem.

1. Warm up. $\triangle ABC \sim \triangle DEF$ with $AB = 6$, $BC = 8$ and $DE = 9$. Find EF.
Write the matching pairs first, then set up one proportion and solve it.

Answer: ________________

2. Find and fix. $\triangle PQR \sim \triangle STU$ with $PQ = 10$, $QR = 14$ and $ST = 15$. Kai writes

$$\frac{ST}{PQ} = \frac{QR}{TU}, \qquad \frac{15}{10} = \frac{14}{TU}, \qquad TU \approx 9.33$$

Kai's two ratios do not run in the same direction. Name the error, then find the correct length of TU.

Answer: __________________

3. Find and fix. $\triangle ABC \sim \triangle FED$ — read that statement carefully. $AB = 12$, $BC = 9$ and $FE = 8$.
Rho pairs $\overline{BC}$ with $\overline{ED}$ correctly but then multiplies by $\frac{12}{8}$ and reports $ED = 13.5$. Explain why a factor greater than 1 cannot be right here, then find ED.

Answer: __________________

4. Two similar triangles have corresponding sides in the ratio $5 : 8$. The smaller triangle has area 50 square centimetres.
Rae multiplies by $\frac{8}{5}$ and reports an area of 80 square centimetres. Name what Rae scaled incorrectly, then find the correct area of the larger triangle, in square centimetres.

Answer: ______________

5. Find and fix. A person 5 ft tall casts a shadow 4 ft long. At the same moment a flagpole casts a shadow 22 ft long. The person and the flagpole with their shadows form similar right triangles.
Ines writes $\dfrac{5}{4} = \dfrac{22}{h}$ and reports $h \approx 17.6$ ft.
State which two lengths Ines paired that do not correspond, then find the correct height h of the flagpole, in feet.

Answer: ______________

6. Challenge. $\triangle JKL \sim \triangle MNP$ with $JK = 15$, $MN = 9$ and $KL = 20$. Tomas writes: "the scale factor is $\frac{15}{9}$, so $NP = 20 \times \frac{15}{9} \approx 33.33$."

(a) Say which triangle is larger and explain why Tomas's answer is impossible before any arithmetic is checked.

(b) Write the rule you would give Tomas for deciding which way up to write the scale factor.

(c) Find the correct length of NP.

Answer: ________________